INCH-POUND

MIL-P-43027E
31 December 1990
SUPERSEDING
MIL-P-43027D
14 September 1976

MILITARY SPECIFICATION

PAPER, CHART, MAP AND BOOK, LITHOGRAPHIC FINISH, COATED AND UNCOATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers four types of lithographic finish papers for use in printing charts and maps and reflects requirements parallel to the Government Paper Specification Standards of the Joint Committee on Printing.

1.2 Classification. The paper shall be of the following types and grades, as specified (see 6.2).

Type I	- High Wet Strength Lithographic Map (JCP E20)
Type II	- Lithographic Finish Map Paper
Grade A	- Deleted
Grade B	- Chemical Wood Map, Lithographic-Finish (JCP E40)
Type III	- Litho Coated Book (JCP A205)
Type IV	- Offset Book Map, Lithographic-Finish (JCP E30)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 9310

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SPECIFICATION

MILITARY

MIL-P-3447 - Paper, Map (Except Diazo-Sensitized), Packaging and Packing Of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-147 - Palletized Unit Loads

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

Government Paper Specification Standards

Published by the Joint Committee on Printing (JCP)

JCP A205 - Litho Coated Book
JCP E20 - High Wet Strength Map, Lithographic-Finish
JCP E30 - Offset Map, Lithographic-Finish
JCP E40 - Chemical Wood Map, Lithographic-Finish
JCP Part II - Testing Standards and Definitions

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

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3.2 Standard sample. The color, finish, formation, and cleanliness of the paper must conform to the standard sample as adopted by the Joint Committee on Printing and as furnished by the contracting officer.

3.3 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.4 Type I paper. Type I paper shall conform to all of the requirements for E20, High Wet Strength Map, Lithographic-Finish set forth in the Government Paper Specification Standards.

3.5 Type II, grade B paper. Type II, grade B paper shall conform to all of the requirements of E40, Chemical Wood Map, Lithographic-Finish set forth in the Government Paper Specification Standards.

3.6 Type III paper. Type III paper shall conform to the requirements of A205 Litho Coated Book set forth in the Government Paper Specification Standards.

3.7 Type IV paper. Type IV paper shall conform to the requirements of E30 Offset Map, Lithographic-Finish set forth in the Government Paper Specification Standards.

3.8 Form. The map paper shall be furnished in sheet or roll form, as specified (see 6.2).

3.8.1 Rolls. In addition to the requirements specified in the Joint Committee on Printing Specification Standards, rolls shall be furnished on a nonreturnable substantial fiberboard core having an inside diameter of 3 inches ± 1/16 inch. The core length shall be not less than the width of the roll material nor greater by more than 1/4 inch. The core shall be plugged at both ends with tightly fitted plugs. The roll diameter shall be as specified (see 6.2) with a tolerance of minus 1 inch or plus 1/2 inch. The rolls shall be suitably restrained to prevent unwinding.

3.8.2 Sheets. Sheet size shall be as specified (see 6.2). Size and tolerance shall conform to the requirements of the Joint Committee on Printing Specification Standards.

3.9 Workmanship. The end item shall conform to the quality of product established by this specification, and the occurrence of defects shall not exceed the applicable acceptable quality levels.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2 and tested as specified in 4.4.3.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 End item examination. The end item shall be examined for the defects set forth in the applicable subparagraphs. The lot shall consist of all material of one type, grade, weight, and size offered for inspection at one time.

4.4.2.1 End item visual examination. The end item shall be examined for the defects listed in table I. The lot size shall be expressed in units of reams or rolls of paper. The sample unit shall be one sheet taken from each ream in the sample or one yard of material taken from each roll in the sample of the full width of the roll, as applicable. No sample unit shall be taken from the first or last convolution of a roll. Both sides of the sample unit shall be examined. The inspection level for appearance shall be I and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5. For workmanship, the inspection level shall be III and the AQL, expressed in terms of defects per hundred units, shall be 0.065.

TABLE I. End item visual defects

Examine	Defect
Appearance	Presence of dirt, oil spots, slivers, shives, scales, shiners, scabs, or mottles larger than the standard sample
	Presence of dirt, oil spots, slivers, shives, scales, shiners, scabs or mottles, more in number than the standard sample
	Finish or formation, as applicable, not equal to or better than the standard sample
	Color not uniform on both sides
	Color not equal to the standard sample
Workmanship	Edges not trimmed clean
	Edges ragged, torn, uneven, or crusted
	Paper curls or does not lie flat (i.e., not flat after reasonable flexing)

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TABLE I. End item visual defects (cont'd)

Examine	Defect
Workmanship (cont'd)	Any hole, tear, cut, wrinkle, fold, chafe mark, scuff mark, slime spot, lump, corrugation, or dog ear causing damage to equipment Presence of lint, fuzz, or any particles that might pick, lift, fluff, or pile on press blanket Any foreign or undigested matter Splice within a sheet or a partial sheet

4.4.2.2 End item examination for defects in roll construction. The end item, when in roll form, shall be examined for the defects listed in table II. The lot size shall be expressed in rolls of paper. The sample unit shall be one roll of paper. Examination shall be made by unwinding the roll. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

TABLE II. End item roll construction defects

Examine	Defect
Assembly of roll	Not suitably restrained to prevent unwinding Material not wound evenly and tightly on roll causing soft or uneven edges or telescoping of roll Material not wound on substantial core; core broken, collapsed, crushed, or mutilated Plugs omitted from one or both ends; improper fit

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TABLE II. End item roll construction on defects (cont'd)

Examine	Defect
Unwinding of roll	Paper slips over core when unwinding Splices comes apart during unwinding More than three splices per roll Splice not evenly and neatly made; does not cover entire width of material Splice taped Splice not flagged with colored marker
Workmanship	Any hole, tear, cut, crease, wrinkle, fold, chafe mark, scuff mark, slime spot, lump, or corrugation Edges not trimmed clean; ragged, torn, uneven, or crushed edges

4.4.2.3 End item dimensional examination. The end items shall be examined for defects listed in table III. The lot size shall be expressed in units of reams or rolls. The sample unit shall be one sheet taken from each ream in the sample or one roll, as applicable. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.0.

TABLE III. End item dimensional defects

Examine	Defect
Sheets:	
Sheet size	Less than specified in either direction
Sheets 8-1/2 by 14 inches or less	Oversize by more than 1/16 inch in either dimension from size specified
Sheets over 8-1/2 by 14 inches	Oversize by more than 1/8 inch in either dimension from size specified

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TABLE III. End item dimensional defects (cont'd)

Examine	Defect
Sheets (cont'd)	
Squareness	Dimensional variation exceeds 1/32 inch for sheets 32 inches or less in length, or exceeds 1/16 inch for sheets of over 32 inches in length
Rolls:	
Width	Less than specified Oversize by more than 1/8 inch
Diameter	More than 1 inch less than specified Over 1/2 inch more than specified
Core	Length less than width of roll or greater by more than 1/4 inch Inside diameter less than 2-15/16 inches or greater than 3-1/16 inches

4.4.2.4 End item count (sheets only) examination. The lot shall be unacceptable if the average count per package of quantity between markers, when paper is skid packed, is less than specified. The inspection level shall be S-2.

4.4.2.5 End item examination for weight per roll. The lot shall be unacceptable if the average weight per roll including nonreturnable core, plugs, and wrapper for all rolls examined is less than indicated. The inspection level shall be S-2.

4.4.2.6 End item examination for color uniformity. The end items shall be examined for the defect listed below. The lot size shall be expressed in terms of shipping containers or rolls. The sample unit shall be one shipping container or roll, as applicable. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Color uniformity	Color not uniform throughout the shipping container or roll, as applicable

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4.4.3 End item testing. The paper shall be tested in accordance with Part II of the Government Paper Specification Standards for conformance to the requirements specified in 3.4 through 3.7. The lot size for the purpose of determining the sample size shall be expressed in units of reams or rolls of one type and size of map paper. The sample unit shall consist of a sufficient number of sheets from each ream or yards of paper from each roll in the sample to provide not less than 1000 square inches of paper. Determinations for moisture content as performed by the contractor shall be made after the paper is trimmed and just prior to wrapping the reams in skid form or on rolls, as applicable. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.4.4 Packaging inspection. The sampling and inspection of the preservation, packaging, and container marking for rolls or sheets, as applicable, shall be in accordance with MIL-P-3447.

4.4.5 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A preservation. Map paper sheets of one type, grade, and size only, shall be preserved in accordance with the applicable requirements of MIL-P-3447. Preservation of rolls is not applicable.

5.1.2 Commercial preservation. Map sheets of one type only shall be preserved in accordance with the applicable requirements of MIL-P-3447. Preservation of rolls is not applicable.

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5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Levels A and B packing. Map paper sheets of one type and size only or each roll of map paper, as applicable, shall be packed in accordance with MIL-P-3447.

5.2.2 Commercial packing. Map paper sheets or rolls of map paper, as applicable, shall be packed in accordance with the applicable requirements of MIL-P-3447.

5.3 Palletization. When specified (see 6.2), paper, packed as specified in 5.2.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, shipments shall be marked in accordance with the requirements of MIL-P-3447.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use.

6.1.1 Type I. Type I papers are intended to be used for high quality multicolor letterpress and offset printing of field books, maps, folded inserts, etc., requiring high wet strength and water-resistant properties and good folding endurance. Printed matter may include text, line illustrations, and halftones. Not suitable for halftone production by letterpress printing.

6.1.2 Type II, grade B. Type II, grade B papers are intended to be used for high quality multicolor letterpress and offset printing of maps, folded inserts, etc., requiring good folding endurance. Printed matter may include text, line illustrations, and halftones. This type is not suitable for halftone production by letterpress printing.

6.1.3 Type III. Type III papers are intended to be used for multicolor face and back offset printing, principally of maps and charts. Printed matter may include text, line illustrations, and halftones up to 150-line screen.

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6.1.4 Type IV. Type IV papers are intended to be used for multicolor letter press and offset printing of maps, folded inserts, etc. Printed matter may include text, line illustrations, and halftones. This type is not suitable for halftone production by letterpress printing.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type, grade, and form required (see 1.2 and 3.8).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
d. When first article is required (see 3.1, 4.3, and 6.3).
e. Roll diameter, when required (see 3.8.1).
f. Sheet size, when required (see 3.8.2).
g. Levels of preservation and packing (see 5.1 and 5.2).
h. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Samples. For access to samples, address the contracting activity issuing the invitation for bids.

6.5 Subject term (key word) listing.

Cartography
Letterpress
Offset
Printing

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:

Army - GL
Air Force - 99

Review activities:

Air Force - 84
DLA - GS

User activities:

Army - ME
Navy - MC, SA

Preparing activity:

Army - GL

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-43027E	1990 December 31

3. DOCUMENT TITLE

Paper, Chart, Map And Book, Lithographic Finish, Coated And Uncoated

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON	c. DATE SUBMITTED (YYMMDD)
c. ADDRESS (Include Zip Code)		

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
U.S. Army Natick RD&E Center	508-651-5221	256-5221
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick [illegible]	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: [illegible]	

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